

NAMMA CLINIC

Digital Healthcare & Clinic Management Platform — Client Application Reference

Platform: Integrated Primary Healthcare Network	Target Audience: Health Administrators & Clinical Staff
Execution Environment: Local Laptop / Clinic Workstation	Date: October 2026 (Phase 28B-0 Approved)

1. Executive Overview

The **Namma Clinic Platform** is a purpose-built, role-governed healthcare operations suite designed specifically for urban primary health centers (PHCs, Namma Clinics, and Urban Health Centers). It provides end-to-end digital lifecycle governance covering citizen intake, vital signs triage, clinician examinations, laboratory investigation processing, electronic prescribing, and pharmaceutical dispensing with First-Expired, First-Out (FEFO) stock control.

2. High-Level Architecture

- The system follows a three-tier local architecture optimized for high performance and offline resilience on local clinic workstations:
- **Presentation Layer:** React 18, TypeScript, and Tailwind CSS client shell delivering dedicated workstations for each role.
 - **Application & API Layer:** Django 4.2 LTS and Django REST Framework providing authoritative RBAC enforcement and facility-scoped querying.
 - **Data Layer:** PostgreSQL 16 relational engine enforcing referential integrity, double-entry inventory movements, and audit logging.

3. Role-Based Workstations

3.1 District Health Officer (DHO)

Responsible for regional healthcare network governance, monitoring public health disease incidence, tracking facility quality indicators, managing staff postings across clinics, and supervising ARS development fund utilization.

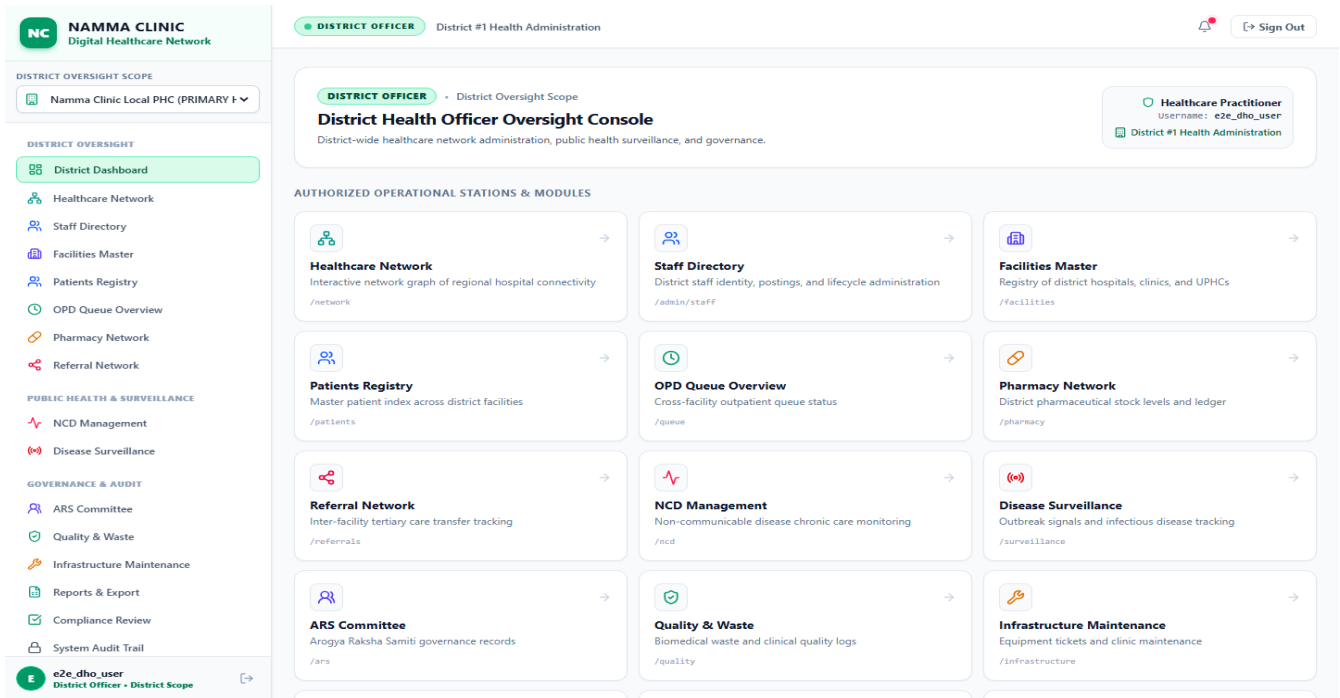


Figure 1 — District Health Officer Dashboard — District-wide overview of patient volumes, clinic uptime, and regional health indicators.

3.2 Hospital / Clinic Administrator

Manages clinic facility operations, staff directory lifecycle (onboarding, operational role assignments, transfers, suspensions), and facility operational throughput.

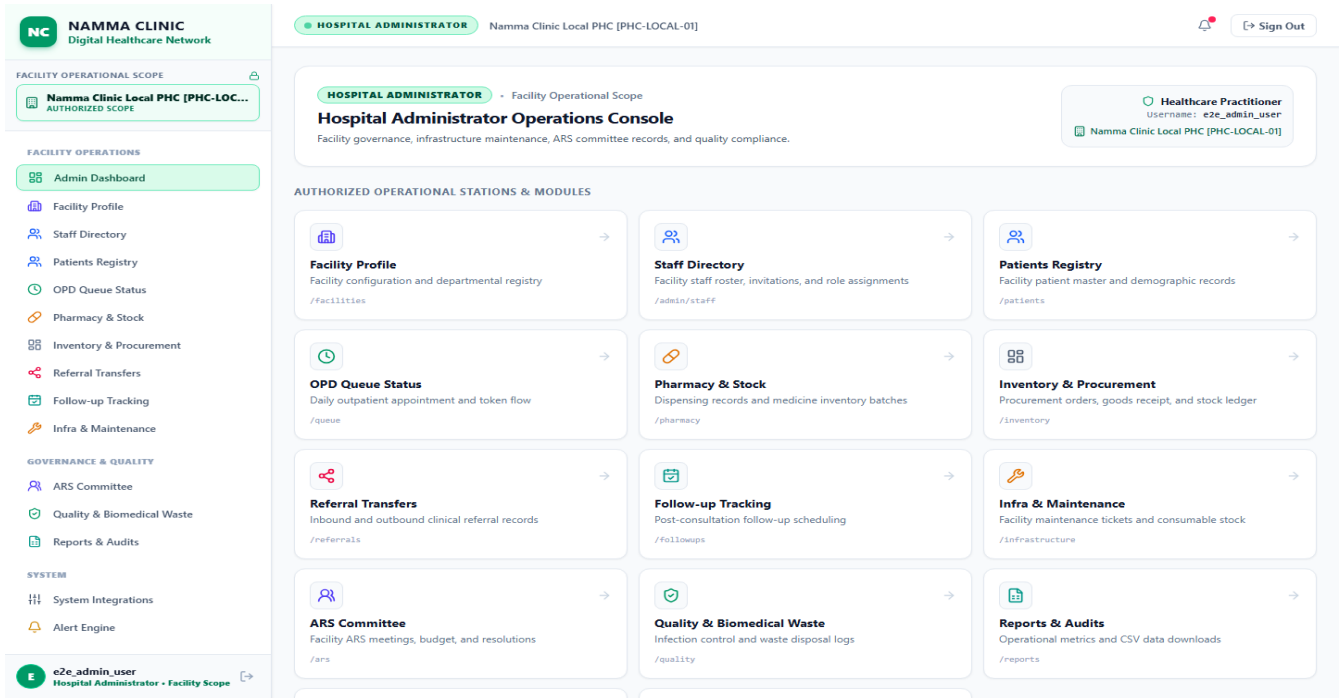


Figure 2 — Hospital Administrator Dashboard — Facility operational KPIs, doctor availability, queue occupancy, and stock alerts.

3.3 Medical Officer (Doctor)

Conducts outpatient consultations, reviews nurse triage vitals, documents formal diagnoses, requisitions diagnostic laboratory tests, and writes structured electronic prescriptions.

NC

NAMMA CLINIC

Digital Healthcare Network

FACILITY OPERATIONAL SCOPE

Namma Clinic Local PHC [PHC-LOC...]

AUTHORIZED SCOPE

CLINICAL CARE

Doctor Dashboard

Patients

Consultation Queue

Doctor Consultation

Diagnostic Orders

Referral Network

Follow-up Care

NCD Longitudinal Care

SYSTEM

Clinical Alerts

DOCTOR

Namma Clinic Local PHC [PHC-LOCAL-01]

Sign Out

Doctor Dashboard

Clinical Encounter & Consultation

Active Encounter: Token #1 - Phase27-5 Citizen b68a92 (WAITING_FOR_TRIAGE)

Phase27-5 Citizen b68a92

FEMALE • 30 yrs

Token #1

Encounter Status: WAITING_FOR_TRIAGE

Queue: TRIAGE

Patient ID: PAT-20261001-5AAEC1

Visit ID: VIS-20261001-F04AEA

Mobile: 98800b808c

Priority: NORMAL

Pre-Consultation Nursing Triage Vitals

Read-only clinical observation (Authorized by Staff Nurse)

Triage vitals have not been recorded for this encounter yet. Doctor may proceed directly with clinical examination.

Patient Longitudinal History (15 previous records)

Visit #155 • Fever, unspecified / Acute viral syndrome

Complaint: High fever for 3 days with generalized weakness and mild headache

2026-09-30

Visit #158 • Type 2 Diabetes Mellitus / Essential Hypertension

Complaint: Acute respiratory tract infection requiring Cefixime

2026-09-30

Visit #157 • Tension Headache

Complaint: Headache & Body Pain

2026-09-30

Medical Officer Consultation Documentation

Chief Complaint *
e.g. High fever with chills, body aches for 3 days

History of Present Illness & Comorbidities
e.g. Known hypertensive, no history of cough or rash

Clinical Assessment & Physical Examination Findings
e.g. Mild pharyngeal congestion, chest clear, no hepatosplenomegaly

e2e_doctor_user

Doctor • Facility Scope

Figure 3 — Doctor Consultation & E-Prescription Form — Comprehensive clinical recording showing diagnosis, vitals review, and e-prescription inputs.

3.4 Staff Nurse

Conducts frontline patient triage, measures vital signs (BP, pulse, SpO2, temperature, respiratory rate), calculates derived BMI, and assigns clinical acuity levels to prioritize urgent care.

NC

NAMMA CLINIC

Digital Healthcare Network

FACILITY OPERATIONAL SCOPE

Namma Clinic Local PHC [PHC-LOC...]

AUTHORIZED SCOPE

PRIMARY CARE & TRIAGE

Nurse Dashboard

Patient Intake

Triage Queue

Nurse Vitals Triage

Follow-up Tracking

NCD Screening

COMMUNITY HEALTH

Outreach & Camps

Wellness Sessions

SYSTEM

Operational Alerts

NURSE

Namma Clinic Local PHC [PHC-LOCAL-01]

Sign Out

Nurse Station Console

Vitals & Clinical Triage

Nursing Triage & Vitals Assessment

Active Encounter: Token #1 - Patient #159 (WAITING_FOR_TRIAGE)

Phase27-5 Citizen b68a92

FEMALE • 30 yrs

DAILY TOKEN #1

PRIORITY NORMAL

Visit ID: VIS-20261001-F04AEA

Phone: 98800b808c

Address: 123 Regression Road, Malleshwaram

Encounter Date: 2026-10-01

Status: WAITING_FOR_TRIAGE

Triage Vital Signs & Clinical Intake Assessment

Authoritative clinical measurement. Values will be evaluated against backend health protocols and forwarded to the Doctor.

Staff Nurse: e2e_nurse_user

Systolic BP (mmHg) *
120
Target < 140

Diastolic BP (mmHg) *
80
Target < 90

Pulse Rate (bpm) *
76
60 - 100

Temperature (°F) *
98.6
Normal 98.6°F

SpO2 Oxygen (%) *
99
Normal ≥ 95%

Resp. Rate (/min)
18
Normal 12 - 20

Height (cm) *
165
30 - 250 cm

Weight (kg) *
65
1 - 300 kg

Computed BMI:
Mathematical derivation (weight / height²) — not a diagnosis
23.9
kg/m²
(derived)

Random Blood Glucose (mg/dL) *
110
< 160 mg/dL

ENCOUNTER INTAKE ESCALATION:
☐ Mark as Emergency ☐ Flag for NCD Intake

Nurse Clinical Observations & Intake Notes

Record patient general condition, consciousness, acute symptoms, mobility, or special intake remarks...

On submission, backend evaluates vitals and transitions visit to Doctor queue.

Save Triage & Send to Doctor

e2e_nurse_user

Nurse • Facility Scope

Figure 4 — Nurse Vitals Triage Form — Vitals capture form with automated BMI computation and clinical acuity prioritization.

3.5 Front Desk Officer

Handles reception intake, searches the demographic registry, registers new citizens with address and vulnerability tags, and generates sequential daily OPD tokens. Front desk officers are strictly isolated from clinical consultation and diagnostic records.

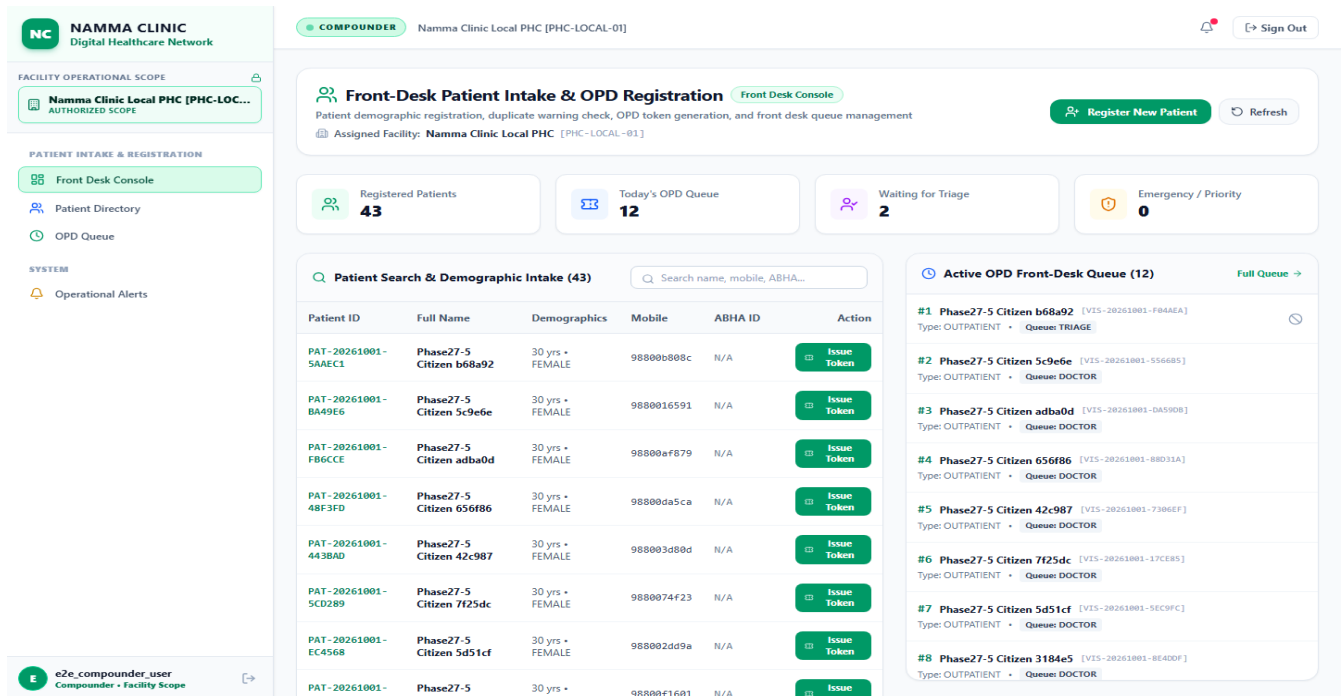


Figure 5 — Front Desk Intake Console — Reception workstation displaying intake metrics, quick demographic search, and daily OPD token stats.

3.6 Diagnostic Lab Technician

Manages the diagnostic testing queue, collects laboratory specimens, inputs investigation values, and issues verified diagnostic reports.

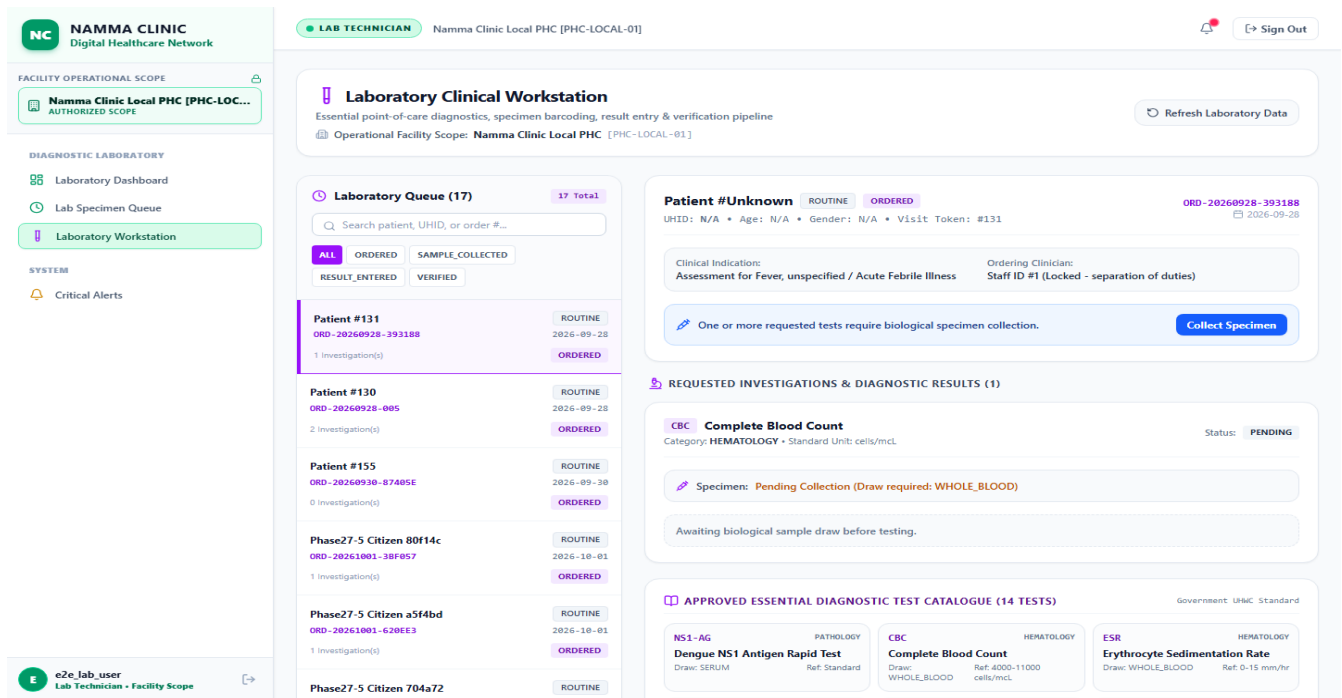


Figure 6 — Diagnostic Investigation Workstation — Diagnostic specimen intake table allowing technicians to verify samples and publish results.

3.7 Pharmacist

Verifies doctor electronic prescriptions, reviews automated FEFO (First-Expired, First-Out) batch selections, dispenses medications, and tracks facility pharmacy stock.

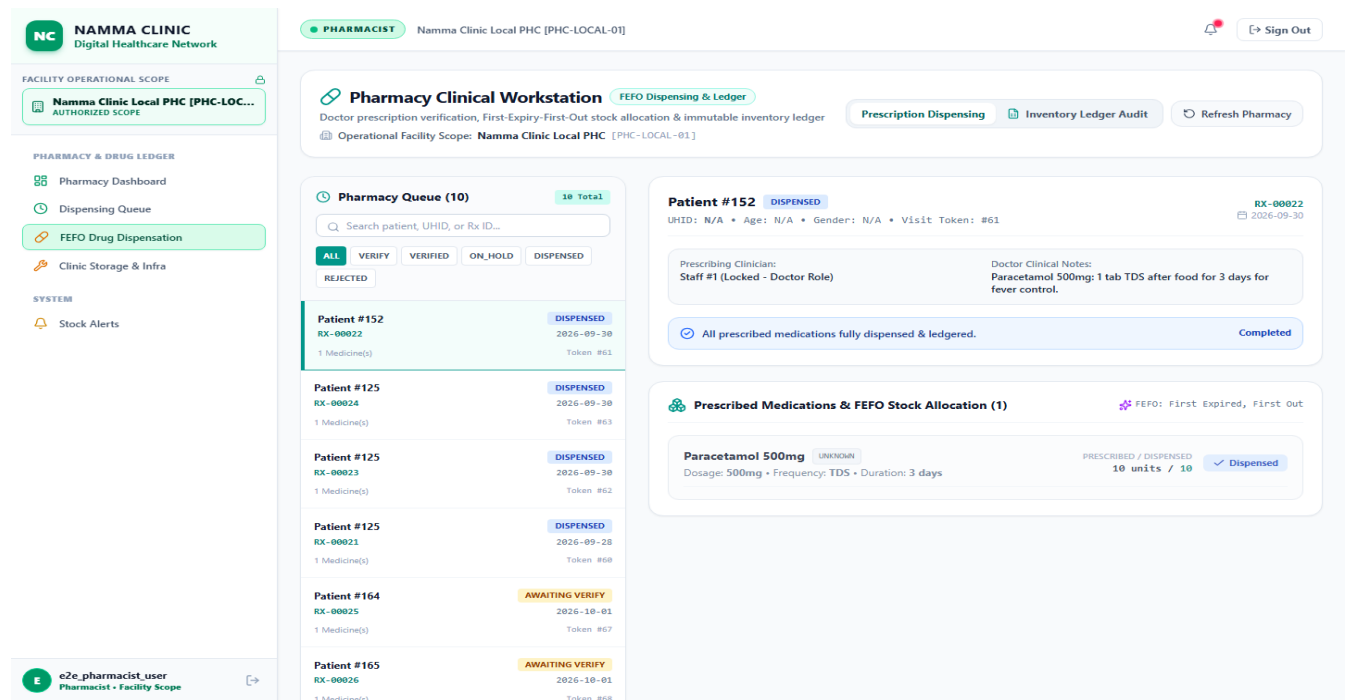


Figure 7 — Pharmacy Dispensing Workstation — Prescription verification table showing doctor orders, recommended FEFO batches, and dispense actions.

3.8 Inventory Officer

Manages facility pharmaceutical procurement, creates supplier purchase orders, processes Goods Receipt Notes (GRN), and reconciles physical warehouse stock balances.

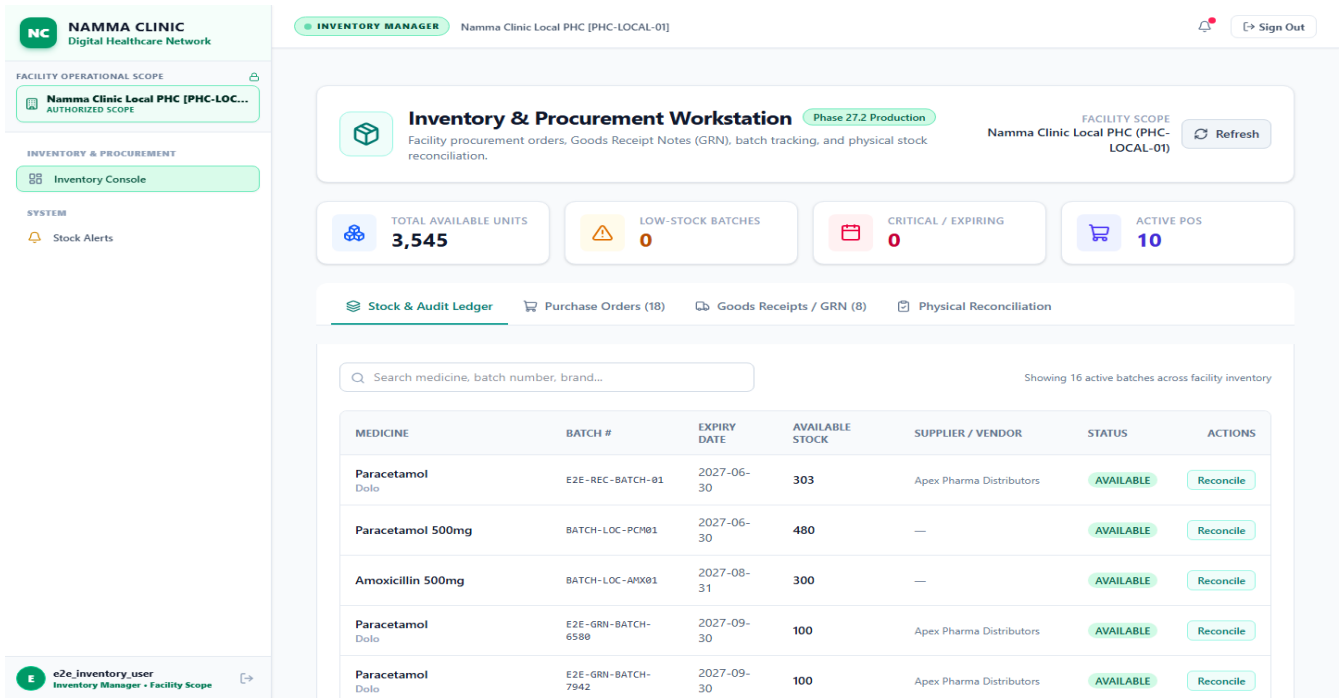


Figure 8 — Inventory Workstation & Batch Balances — Real-time facility inventory ledger displaying batch quantities, expiry dates, and unit prices.

3.9 Dual-Role Configuration (Small-Clinic Rule)

Provides operational flexibility for small clinics where a single staff member fulfills multiple duties (e.g., Nurse + Front Desk Officer or Pharmacist + Inventory) as separate authoritative role records without creating hybrid roles.

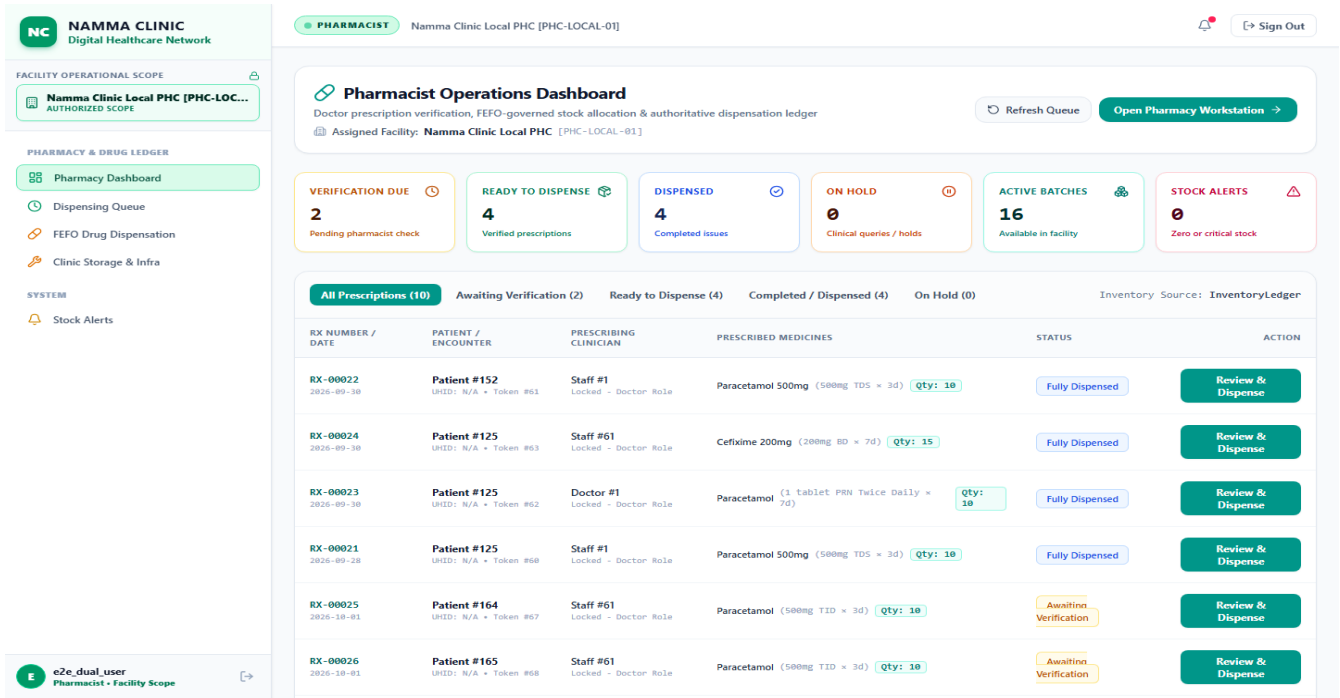


Figure 9 — Dual-Role Pharmacy & Procurement Console — Workstation providing unified access to both dispensary and inventory operations.

4. Patient Journey & Workflow Stages

The patient journey moves sequentially through five auditable stages: Citizen Registration → OPD Token Issuance → Nurse Vitals Triage → Doctor Consultation & Prescription → Diagnostic Lab / Pharmacy Dispensing.

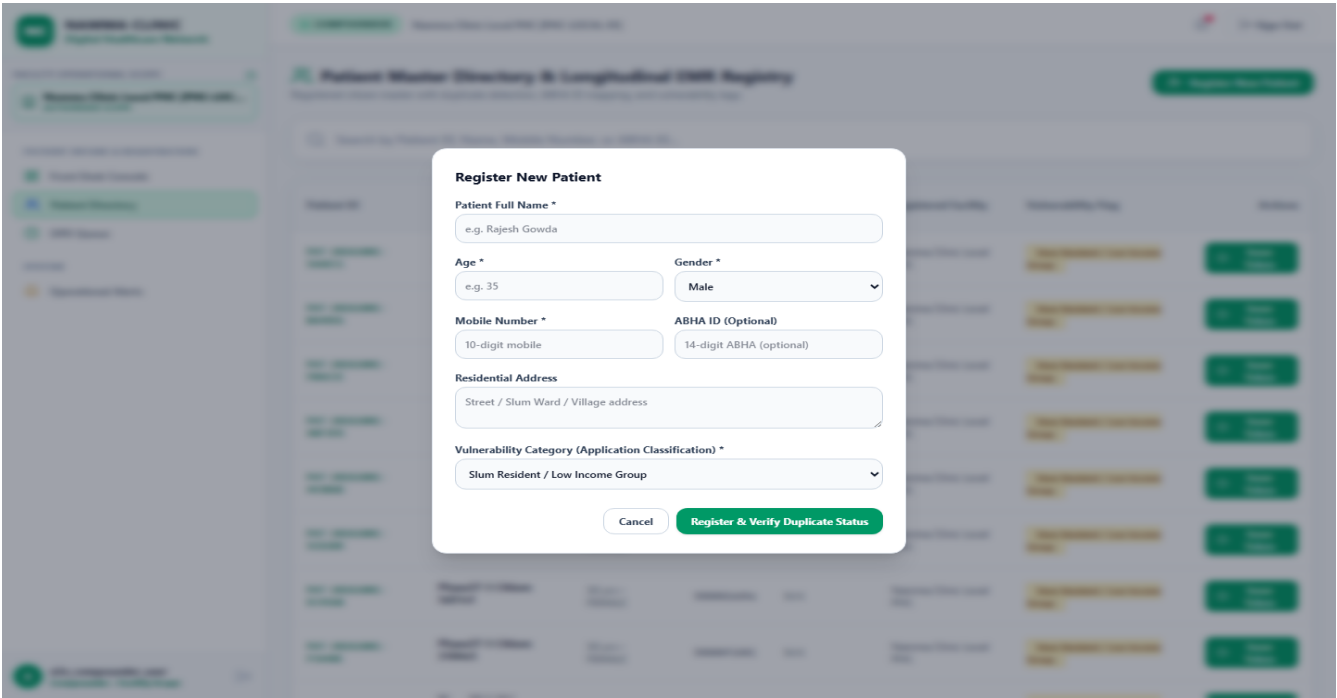


Figure 10 — Citizen Registration Modal — Front Desk Officer registers newly presenting citizens with demographic and emergency details.

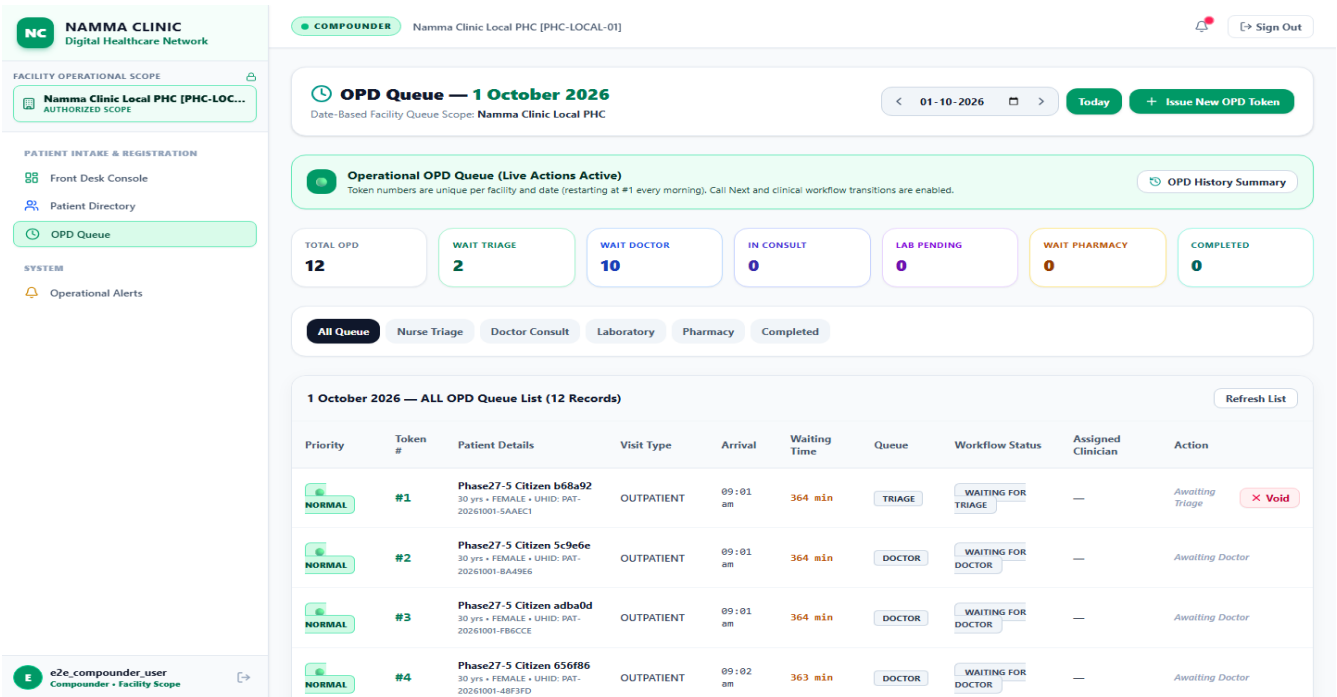


Figure 11 — OPD Token & Waiting Queue — Live queue displaying sequential tokens, patient names, and priority classifications.

5. Inventory & Pharmacy Separation of Duties

Warehouse procurement and clinical medication dispensing are architecturally isolated. Storekeepers cannot dispense medications, and pharmacists cannot directly mutate warehouse stock. All balance updates flow through an immutable double-entry movement ledger.

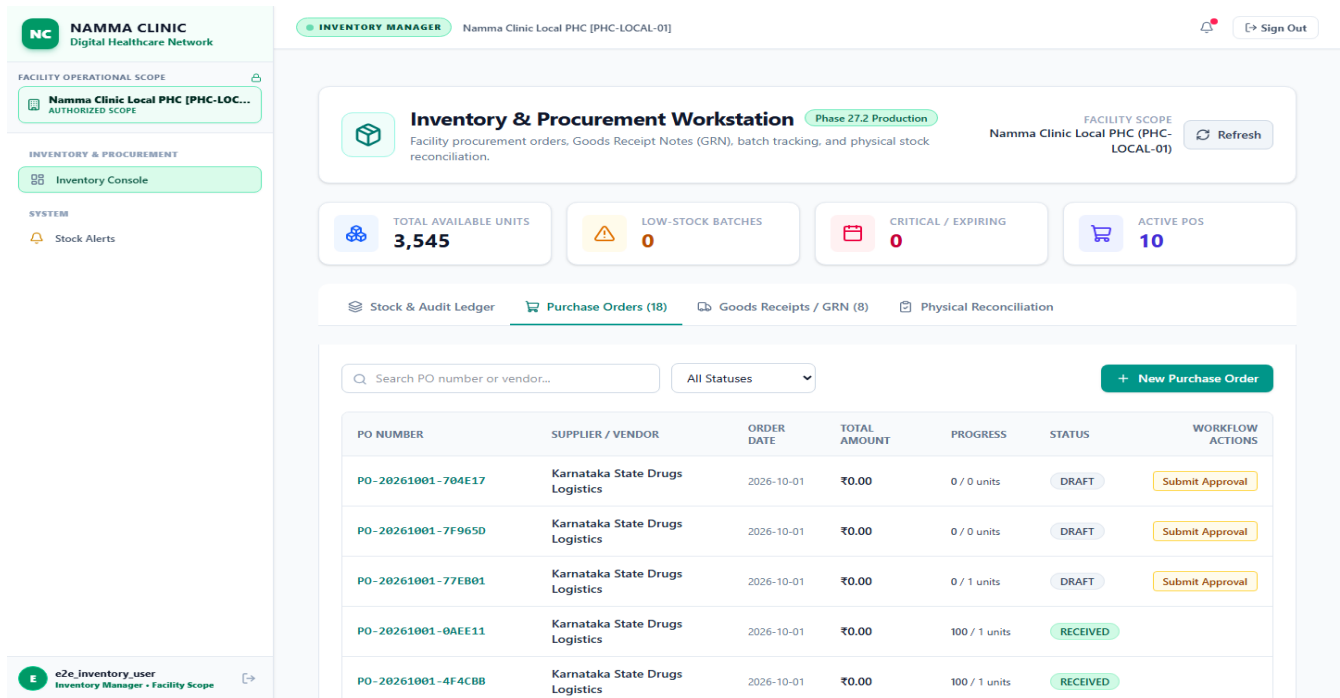


Figure 12 — Warehouse Purchase Orders Tab — Procurement table managing external supplier orders and Goods Receipt Notes.

6. Security & Role-Based Access Control (RBAC)

Every client route and API endpoint is strictly protected by authoritative backend role and facility-scoping checks. Unauthorized access attempts are intercepted by clean security cards and logged to the tamper-evident audit trail.

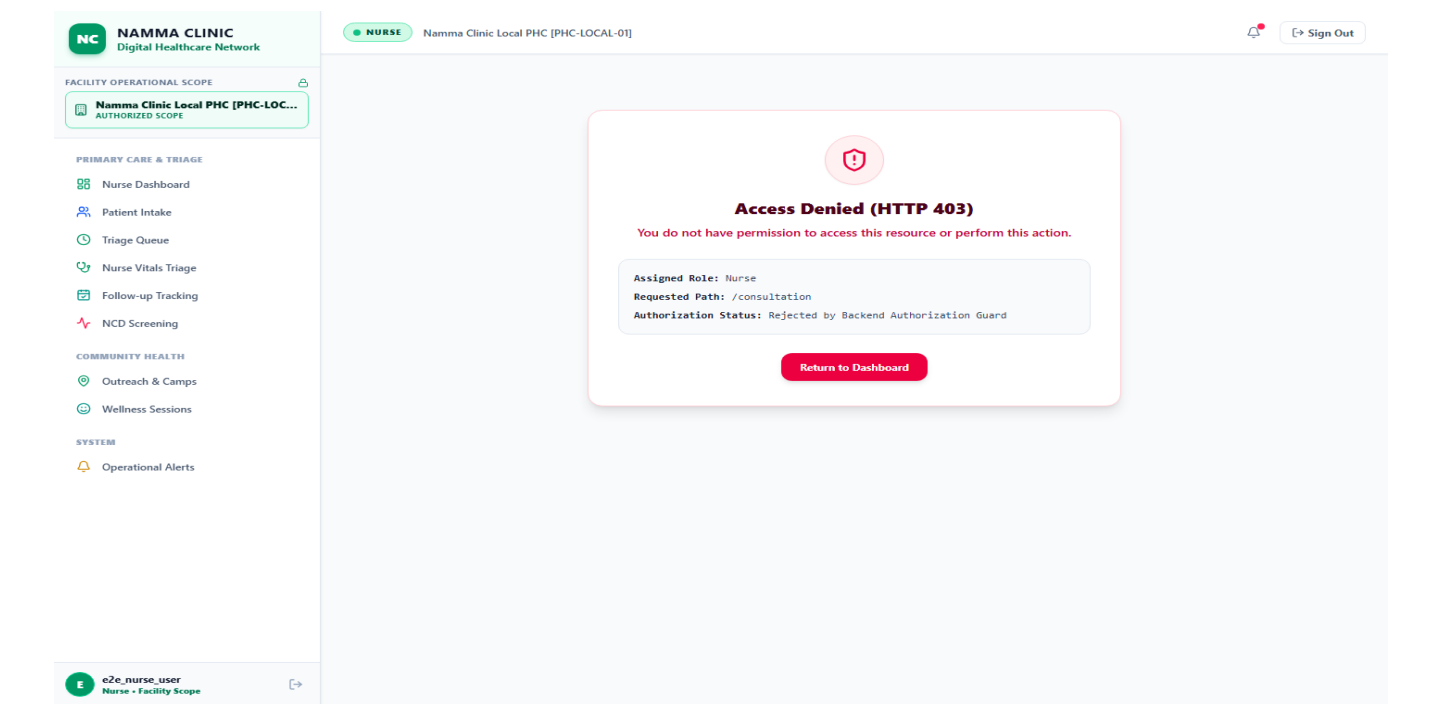


Figure 13 — Security Boundary Enforcement — Clean Forbidden Card displayed when an operational role attempts to navigate outside its authorized boundary.

7. Current Application Scope

Capability / Domain	Implementation Status	Operational Scope
Authentication & RBAC	Fully Implemented	8 canonical roles, JWT auth, facility scoping
Front Desk Intake & Tokens	Fully Implemented	Citizen registration, deduplication, token queue
Nurse Vitals & Triage	Fully Implemented	Vitals entry, BMI calculation, acuity scoring
Doctor Consultation & E-Rx	Fully Implemented	Diagnoses, clinical notes, e-prescriptions, lab orders
Laboratory Diagnostics	Fully Implemented	Diagnostic queue, specimen intake, result publishing
Pharmacy & FEFO Dispensing	Fully Implemented	Rx verification, hold notes, FEFO stock deduction
Warehouse Inventory	Fully Implemented	Purchase orders, GRN, double-entry movement ledger
Staff Lifecycle Admin	Fully Implemented	Staff directory, role assignments, transfers
Local Laptop Execution	Active Standard	PostgreSQL 16, Django 4.2 LTS, React 18 / Vite
External Integrations	Readiness Mode	ABDM/ABHA data models ready; mock endpoints